R9 Radial Engine

Design notes for the nine-cylinder master and articulating rod crank train modelled in Fusion as `R9 Engine Assembly`. They cover the layout decisions, the knuckle pin tuning, crankshaft balance, the structure of the CAD model and the checks run on it.

DOCUMENT	REVISION	STATUS	CAD FILE
R9-DN-001	A	Layout study	R9 Engine Assembly v4
PROJECT	UNITS	ROTATION	
R9 Radial Engine	mm, kg, deg	CW from rear	

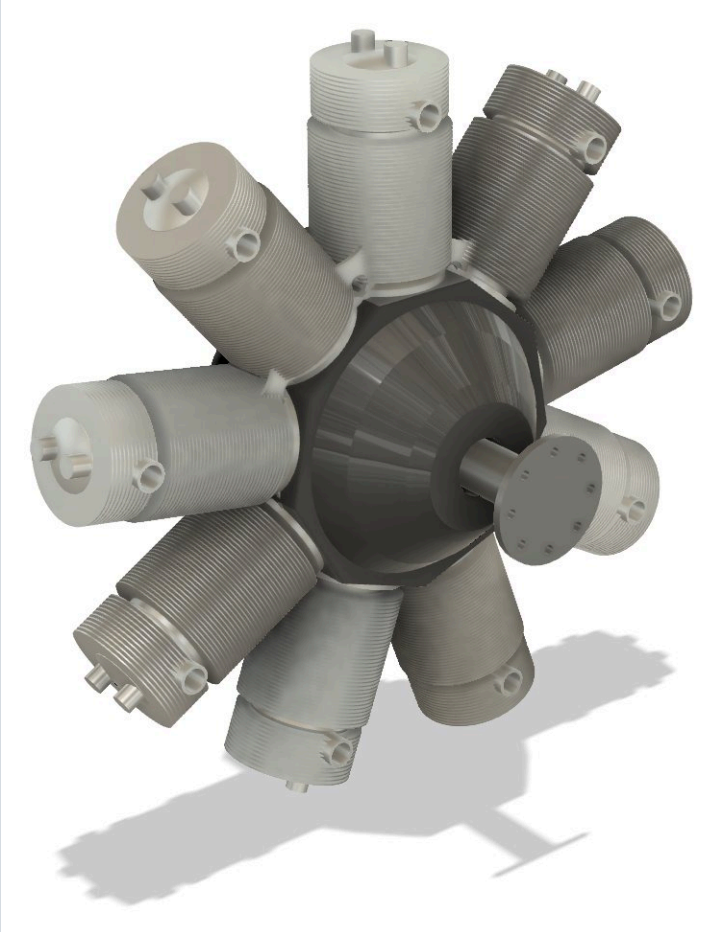


Fig. 1 Assembly, front three-quarter view, crank at 20 deg.

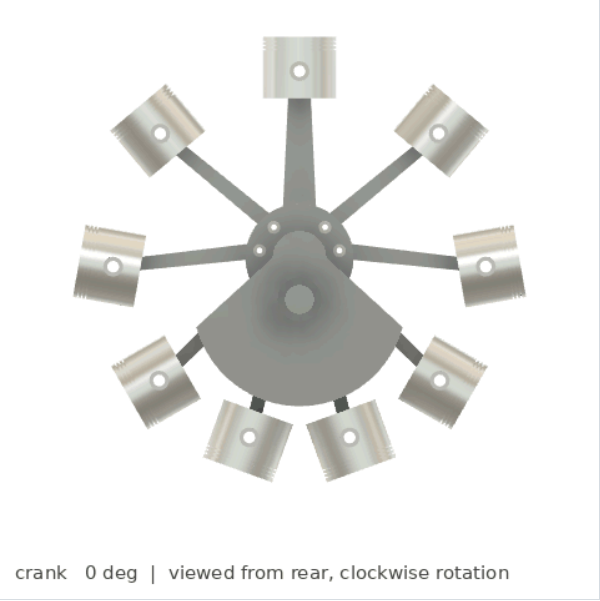


Fig. 2 One revolution driven through the main bearing joint, viewed from the anti-propeller end. Crankcase and cylinders hidden.

1 Specification

ITEM	VALUE	NOTE
Configuration	9 cyl, single row	Air cooled, four stroke, firing order 1-3-5-7-9-2-4-6-8 at 80 deg intervals
Bore x stroke	150 x 170	Stroke/bore 1.13
Swept volume	27.04 L	3.004 L per cylinder
Compression ratio	6.8 : 1	48 mm spherical cap chamber plus 2 mm crown clearance at TDC
Master rod, crankpin to wrist pin	380	r/L = 0.224
Link rods	300	On a nominal 80 mm knuckle pin circle, individually tuned (section 3)
Wrist pin travel from crank axis	295 to 465	Same TDC height on all nine cylinders after tuning
Deck height	537	Crank axis to top of bore
Modelled mass	336 kg	Crank train, cylinders and case only (section 7)

2 Layout and rod length

The rod length was set by packaging rather than by angularity. With nine cylinders on 40 deg spacing, two adjacent 150 mm pistons stop clearing each other once the wrist pin comes inside a radius of about 219 mm, and the skirts sit 55 mm below the pin. A 300 mm master rod put the pistons at BDC too close to the crank axis for the skirts to clear, so the master rod went to 380 mm. That puts the wrist pin at 295 mm at BDC and the skirt bottom at 240 mm, where adjacent pistons have 26.3 mm between them.

The longer rod also lowers the master rod swing to 12.9 deg either side of the cylinder axis, which keeps the piston side load and the angular travel of the link rods relative to the master rod small.

The cylinder hold-down flanges and the inner barrel fins would still meet their neighbours, so each cylinder carries two flats 2 mm off the bisector between cylinders (feature `Fin clearance`). The resulting gap between neighbouring fin stacks is 4 mm.

3 Articulated rod kinematics

Only cylinder 1 runs on a true slider crank. The link rods pivot on knuckle pins carried by the master rod big end, and the big end rocks as the crank turns. With all knuckle pins on one circle at exactly 40 deg spacing, each link-rod cylinder ends up with a slightly different TDC height, a slightly longer stroke than the master cylinder, and a TDC that is offset in crank angle. The TDC height spread matters most, because it sets the compression ratio of each cylinder.

Each knuckle pin was given its own radius `a_k` and angular offset `d_k` from the 40 deg station. These sixteen values were found by least squares on the closed-form solution, sampled at 0.25 deg over a revolution, with weights of 4 on TDC height error, 2 on stroke error, 0.2 on TDC timing and a small penalty keeping the pins near the nominal circle.

```
master:  s1 = r cos(t) + sqrt(L^2 - r^2 sin^2(t)),   beta = atan2(-r sin t, s1 - r cos t)
pin k:   K_k = C + a_k (sin(beta + 40k + d_k), cos(beta + 40k + d_k))
piston:  s_k = K_k.u_k + sqrt(l^2 - |K_k|^2 + (K_k.u_k)^2)
```

KNUCKLE PIN LAYOUT	TDC HEIGHT SPREAD	MAX STROKE	TDC TIMING OFFSET
Equal pins, 80 mm circle, 40 deg spacing	2.40 mm	170.57 mm	±3.50 deg
Tuned pins (as modelled)	<0.001 mm	170.00 mm	±3.75 deg

The tuning trades a quarter of a degree of extra timing offset for equal compression and stroke on every cylinder. The timing offset is fixed for each cylinder, so it would be taken out in the ignition and valve timing.

KNUCKLE PIN POSITIONS ON THE MASTER ROD

CYLINDER	A_K (MM)	STATION (DEG)	D_K (DEG)	DIMENSIONED ANGLE (DEG)
2 and 9	81.0692	40 / 320	−0.0121 / +0.0121	39.9879
3 and 8	82.5993	80 / 280	−0.2702 / +0.2702	79.7298
4 and 7	82.0350	120 / 240	−0.8174 / +0.8174	119.1826
5 and 6	80.3188	160 / 200	−0.5510 / +0.5510	159.4490

The pin set is symmetric about the master rod centreline, which makes the whole mechanism a mirror image of itself. Angles are measured from the master rod centreline in sketch `Master Rod / Pin layout`, where they are held to four decimals.

4 Crankshaft balance

In a single-row radial with three or more cylinders, the primary reciprocating forces add to a vector of constant magnitude $(N/2)\, m_{rec}\, r\, \omega^2$ that turns with the crank. A counterweight can therefore cancel the whole primary force along with the rotating mass at the crankpin, which an in-line engine cannot do.

MASS AT THE CRANKPIN	KG	BASIS
Rotating: master rod big end, knuckle pins, link rod big ends	18.87	Rods split by centre of mass position between their eyes
Reciprocating, cylinder 1	6.07	Piston, pin and master rod small-end share
Reciprocating, cylinders 2 to 9	4.40 each	Piston, pin and link rod small-end share
Primary equivalent, N/2 x mean m_{rec}	20.63	
Required crank unbalance, opposite the pin	3358 kg.mm	(18.87 + 20.63) x 85 mm
Modelled crankshaft	3370 kg.mm	Read from Fusion mass properties

The counterweight is a 150 deg sector whose radius is driven by `CounterweightRadius`. It was converged on the CAD body by a secant search on that parameter and rounded to 217.5 mm, leaving 0.4 % overbalance. The secondary forces and the variation caused by the articulated rods are not balanced; that is normal for this layout and would be handled by mounts and, if needed, a dynamic damper.

5 Cylinder

The cylinder is one body built the way it would be drawn by hand: a revolved blank, one barrel fin and one head fin each patterned along the axis (34 at 7 mm, 12 at 7.5 mm), port bosses, rocker box bosses, then the bore cut up from the base to the deck and the chamber revolved out of the head. The bore diameter, deck height and chamber height all follow user parameters. Intake ports face the rear and exhaust ports face the front.



Fig. 3 Crank train with the crankcase and cylinders hidden.

6 CAD model structure

COMPONENTS

Eight components, each with its own origin on the crank axis or on its own pin centre: Crankcase, Cylinder, Crankshaft, Master Rod, Link Rod, Piston, Wrist Pin and Knuckle Pin. Repeated parts are instances of a single component, so a change to one cylinder or one piston carries to all of them. Cylinder numbers follow the usual convention, clockwise from the rear with number 1 at the top. Link Rod instance `n` belongs to cylinder `n+1`, and Knuckle Pin instance `n` likewise.

TIMELINE

The timeline is grouped by component, followed by `Instances`, the ground, the as-assembled position, `Joints` and `Rigid groups`. Sketches use connected geometry and true arcs. The master rod pin layout, the crank web and the link rod pin layout are fully constrained.

PARAMETERS

PARAMETER	VALUE	DRIVES
`Bore`	`150`	Cylinder bore, chamber, piston diameter
`Stroke, CrankThrow`	`170, 85`	Crankpin position on crank, master rod
`MasterRodLength, LinkRodLength`	`380, 300`	Rod centres
`CompressionHeight, DeckHeight, ChamberHeight`	`70, 537, 48`	Piston crown, bore length, chamber
`CrankpinDia, BigEndAllowance`	`80, 1`	Crankpin, master rod big-end bore
`WristPinDia, KnucklePinDia`	`40, 26`	Pins and their bores in rods and pistons
`PistonClearance`	`0.8`	Piston diameter, diametral
`CounterweightRadius, CounterweightHalfAngle`	`217.5, 75 deg`	Crank web sketch
`CylinderCount, fin counts and pitches`	`9, 34/12, 7/7.5`	Patterns

Stroke, rod lengths and deck height are layout values fixed by the study above. They update the parts, but the joints were made with the parts assembled at crank angle 0, so a change to any of them needs the assembly re-seated and the kinematic check rerun.

JOINTS AND HOW TO DRIVE IT

The crankcase is grounded and locked to the nine cylinders in a rigid group. The crankshaft turns in the case on `Main bearing`; the master rod turns on `Crankpin`; each link rod turns on its knuckle pin and its wrist pin; each piston slides in its bore. The master rod and its knuckle pins form one rigid group and each piston is grouped with its wrist pin. All nine slider-crank loops are closed, so the mechanism has one degree of freedom. Drag the crankshaft, or set the angle of `Main bearing` (positive is clockwise seen from the rear, the normal direction for a tractor engine).

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7 Checks

CHECK	METHOD	RESULT
Joint model against closed-form kinematics	Main bearing swept 0 to 360 deg in 5 deg steps, all nine piston positions read back from the assembly (657 values).	PASS max error 0.00012 mm
Running clearance and interference	Minimum distance between 79 pairs of parts that can meet, at every 5 deg of a full revolution. Pin-in-bore pairs excluded.	PASS no contact; smallest running clearance 4.6 mm (below)
Balance	Crankshaft mass properties against the requirement in section 4	PASS 0.4 % over

MINIMUM RUNNING CLEARANCES

PAIR	MIN (MM)	WHERE
Link rod to master rod slot, axial	1.0	Design side clearance, slot 32 mm, eye 30 mm
Crankshaft journal to case housing	1.0	Radial allowance for the main bearing, not modelled
Fin stack to adjacent cylinder	4.0	Fin clearance flats (static)
Link rod to its own cylinder skirt	4.6	Cylinders 5 and 6 at crank 100 deg and 260 deg. Smallest running clearance in the engine
Master rod to cylinder 1 skirt	7.3	Crank 60 deg and 300 deg, near maximum rod angle
Link rod to link rod	7.6	Cylinders 3 and 4 (and 7 and 8) near the crank
Master rod to crankcase	10.6	Crank 65 deg and 295 deg
Crankshaft to link rods	16.3	
Counterweight to piston skirt	22.5	Piston at BDC
Piston to adjacent piston	26.3	Both near BDC

MASS BREAKDOWN

PART	MATERIAL	EACH (KG)	QTY	TOTAL (KG)
Crankcase	Aluminium 2014-T6	72.14	1	72.14
Cylinder	Aluminium A356-T6	13.02	9	117.14
Crankshaft	Steel 4340	87.02	1	87.02
Master rod	Steel 4340	11.77	1	11.77
Link rod	Steel 4340	2.03	8	16.26
Piston	Aluminium 2014-T6	2.52	9	22.72
Wrist pin	Steel 4142	0.86	9	7.73
Knuckle pin	Steel 4142	0.21	8	1.68
Total modelled				**336.46**

8 Scope and open items

This is a crank train layout. It is complete enough to check kinematics, packaging and balance, and it is not a detailed design. The following are not modelled and are the next items to take on.

- OPEN Valve train: cam ring, tappets, pushrods and rockers. The rocker boxes are bosses only.
- OPEN Bearings: main bearings, big-end shells and knuckle pin bushes are represented by bore allowances.
- OPEN Induction, supercharger, ignition and the accessory drive in the rear case.
- OPEN Propeller reduction gearing. The propeller flange is on the crankshaft.
- OPEN Structural checks: crankpin and web stresses, rod buckling, cylinder hold-down loads. Gas loads have not been applied.
- OPEN Link rod to skirt clearance on cylinders 5 and 6 is 4.6 mm. A skirt relief notch or a slimmer link rod shank near the big eye would give more margin.
- OPEN Ring land and valve pocket volumes, which will pull the compression ratio slightly below 6.8.

R9-DN-001 rev A. Kinematic figures come from the closed-form model; masses and clearances are read from the Fusion model at version 4.